



# Oakwood University AI Studio & Engineering Lab

Turnkey proposal & execute-ready estimate for a 10-seat AI production studio, a hands-on engineering/soldering makerspace, and the department-wide power, network, camera, and access-control infrastructure that protects the investment — orchestrated end-to-end by Island Development Crew.

Prepared: June 2026

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**Provider:** Jonathan Isaac  
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**What this is.** A planning-grade, audit-aware General-Contractor proposal built from an on-site walkthrough (May 26, 2026), **71 verified site photographs**, and current June-2026 vendor pricing with named Huntsville subcontractors. IDC orchestrates every trade under one accountable point of contact, one transparent fee, and one progress dashboard. *Figures are planning ranges; each line is re-quoted to a sealed vendor/sub bid before purchase.*

10-Seat AI Studio · Rm 409

Soldering/Math Lab · Rm 407

Engineering · Rm 305 + 310

UniFi + pfSense Network

UniFi Protect Cameras

Multi-Door Access

≥4 Bambu Lab Printers

2084 / Lennox Curriculum

RECOMMENDED EXECUTION — ALL-IN (IDC-ORCHESTRATED)

# \$172,000

Range \$170,625 – \$198,498. Includes equipment, subbed facility work, GC coordination, contingency, and the 10% IDC orchestration fee. Title III grant-eligible portion ≈ \$93.5K; facility/capital portion ≈ \$60.1K.

**Mobilization deposit \$68,000** initiates procurement & sub scheduling (see Payment Schedule, p.11).

**Secure payment:** [Island Dev Crew Stripe payment link](#)  
(\$68,000 mobilization deposit)

**Signing:** signature-ready PDF with direct blocks (p.11);  
DocuSign / SignWell / Dropbox Sign accepted.

Department-Wide Vision (full robust build incl. dept cabling, dept cameras, Room-405 bathroom & raised floor) is on p.13 as the donor / naming-gift envelope at **\$245K–\$330K**.

## The opportunity

Oakwood does not need another computer lab. It needs an **AI production studio** where Math & Computer Science students *ship* AI — apps, agents, fabricated hardware, and research artifacts — backed by reliable power, network, and security. As the only Seventh-day Adventist HBCU, Oakwood can also give students a credible **responsible-AI ethics** voice (curriculum, p.12).

## Why IDC as GC

One orchestrator manages every vendor and Huntsville subcontractor (electrical, low-voltage, access, GC), holds a single **3–7% coordination markup** on subbed facility work and a transparent **10% orchestration fee**, and keeps faculty & administration informed through a live progress dashboard with weekly/monthly reports.

## 1 · Executive Summary & How to Read This

This package separates the project into the three quote families requested, because each draws on **different funding**. Selling all three as one grant line will be clawed back at audit.

| Track                                | Scope                                                                                | Planning range                | Funding lane                                  |
|--------------------------------------|--------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------|
| Quote 1 — AI Studio + Engineering    | Compute, Bambu fleet, soldering/CNC, AV + 3 boards, furniture, consumables, software | \$88.2K–\$98.8K               | Mostly <b>Title III grant-eligible</b>        |
| Quote 2 — Power / Electrical         | FPE panel replacement, subpanel, dedicated circuits, raceway, raised floor, permits  | \$54.4K–\$61.0K               | <b>Facility — NOT grant</b> (capital / donor) |
| Quote 3 — Network + Cameras + Access | UniFi + pfSense, Protect cameras, 4-door Access; + structured cabling labor          | Equip grant; cabling facility | <b>Mixed</b>                                  |

## 2 · Two Tiers — Recommended vs. Vision

### Recommended Execution — ~\$172,000 all-in

Studio/lean compute, lab-focused facility scope (power + cabling to the labs + MDF; raised floor in the AI Studio only; *no* Room-405 bathroom). The number to commit to — top of the \$150–175K band you targeted; the execute-ready invoice (p.11) bills this.

### Department-Wide Vision — \$245K–\$330K

Full robust transformation: pooled compute (RTX PRO 6000), dept-wide cabling (80 drops + fiber), dept-wide cameras, full electrical, raised floor *plus* the Room-405 bathroom and dept room-mod, 5-door access. The **donor / naming-gift** envelope (p.13).

## 3 · The Funding Firewall (critical)

**Title III** covers **supplies** (<\$2,000/unit) and **equipment** (≥\$2,000/unit) tied to the academic mission, with an audit trail and a **June 30** spend pressure. It does **not** cover **facilities** — electrical service, structural cabling labor, raised floor, construction, HVAC, the bathroom. Quote 2 in full and the cabling-labor / door-install portions of Quote 3 must come from **capital, physical-plant, donor, or match funds**.

## 4 · Sequencing — two hard dependencies + 90-day plan

- **No compute energizes before Quote 2 circuits.** The RTX workstation (575–600W) and DGX nodes need the dedicated 20A/30A circuits first.
- **The studio is unusable before Quote 3 drops.** Room 409 currently has *one* data drop; the 10GbE backbone + new drops come before deployment.

| Phase         | Actions                                                                                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Now → June 30 | Lock education-priced POs for Quote 1 (quotes expire 14–30 days); commission the licensed-electrician load study; campus-IT vendor ruling; confirm tax-exempt + line-by-line grant eligibility. |
| 30 days       | Sealed sub bids (3 each: electrical, low-voltage, access, GC); permits; cabling + raised-floor design; finalize room dimensions & seat count.                                                   |
| 90 days       | Install <b>power</b> → <b>cabling/network</b> → <b>compute</b> → <b>cameras/access</b> → <b>demo day</b> . Display boards + dashboard go live; first cohort onboards.                           |

## 5 · Existing-Conditions Assessment (Walkthrough, May 26 2026)

A true on-site assessment was performed with Dr. Smith and is documented by 71 verified photographs (exhibits, pp.6–7; full set + explanations on the online resource page). Room → lab mapping:

| Room       | Becomes                 | Condition found (drives scope)                                                                                                                                                              |
|------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 409        | AI Studio (10–13 seats) | Preferred room. <b>One data drop</b> ; breakers trip on space heaters; keep dual smartboards (one Wacom) + 2 projectors; fire-egress concern; surface-run power on bare block.              |
| 407        | Math & Soldering Lab    | Convertible; <b>blue steel pegboard workbenches + empty cabinetry + lab sink</b> (reusable); an "L-bend" fits a fume extractor.                                                             |
| 305 + 310  | Engineering Lab         | 305 = project room (benches, existing MakerBot 3D printers). 310 = computer lab with <b>rows of tower desktops (NVIDIA-GPU) to repurpose for AI hosting</b> — saves the engineering budget. |
| MDF closet | Network core + power    | <b>Federal Pacific (FPE) panelboard "2C"</b> with exposed bus + mixed 15/20/25/30A breakers, co-located with the network patch/fiber field (not to code); drops out ~monthly.               |

△ **Priority safety finding:** the MDF closet contains a **Federal Pacific Electric (FPE) panel** (EX-01-03) — a panel type with a documented industry history of breakers failing to trip on overload, a recognized fire hazard. Replacing it is strongly recommended and is carried in Quote 2. Whole-complex: no 90° angles, 15 "Frankensteined" AC units, windows with failed glazing/interpane fogging, bare cinderblock.

### In Dr. Smith's words

"I've only got the one data drop right there." — (asked: in the entire room?) — "Yes, that's it right there."

"These electrical units, they can't handle nothing... I got portable heaters... Boom." → "That means your breakers need to be updated."

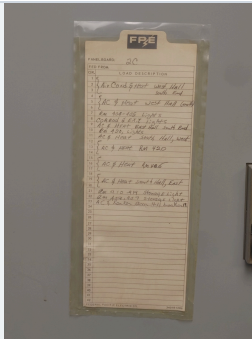
"I know this is not to code. We're not supposed to have a closet next to the fuse box."

"The main objective: secure and always changing... nobody could copy it or retain that information." (access control)

"We want AI. We need power. Power is the number one thing you invest in."

## 6 · Photo Exhibit Addendum — Part 1 of 2 (verified on-site)

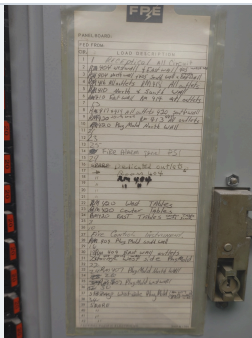
Each photograph was individually reviewed; captions describe what is actually in frame. All 71 photographs at full resolution with explanations: [resources.html](#).



**EX-01 · power · MDF/networking closet:** FPE panelboard 2C directory card, handwritten, showing circuits dominated by AC/heat and lighting loads for the 400-series rooms and hallways (incl. Rm 404-405, 409, 420). The



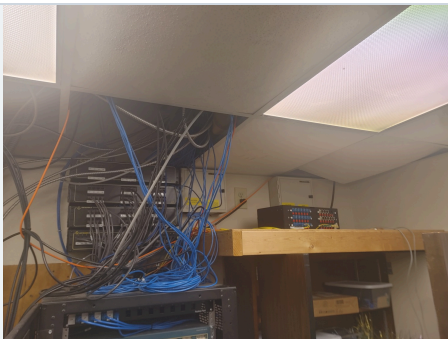
**EX-02 · power · MDF/networking closet:** Interior of FPE panel 2C with deadfront off: exposed bus bars and a mix of 15/20/25/30A single- and double-pole breakers hand-labeled to rooms 405, 409, 420 and hallway/AC zon



**EX-03 · power · MDF/networking closet:** Sharp close-up of the FPE panel directory enumerating receptacle, plug-mold, dedicated-outlet, lab-table, fire-alarm, and '3D'-printer circuits across Rooms 404-420, with only



**EX-04 · data-network · Hallway / classroom wall:** Abandoned/exposed low-voltage drop: a surface-mount junction box with a loose twisted-pair (blue/yellow) hanging free and tucked behind the baseboard, not terminated in a prop



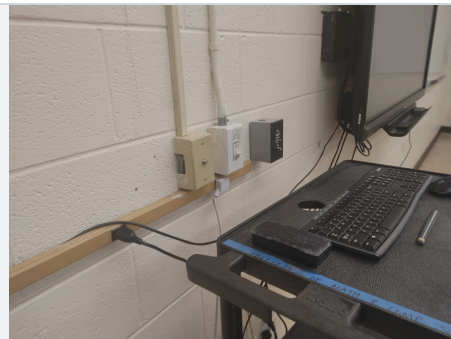
**EX-05 · data-network · MDF/networking closet:** MDF closet overview: a wall patch field and fiber breakout panel buried under unmanaged blue/gray/orange cabling routed through an open ceiling-tile gap, mounted on a bare wood



**EX-06 · data-network · MDF/networking closet:** Detail of the closet fiber distribution panel (rows A/B/C with red/blue couplers and yellow jumpers) beside a fiber-optic caution label and lone wall outlet, all on bare lumbe



**EX-07 · AV · Classroom (AI Studio candidate):** Primary AV wall: large interactive flat-panel display on an articulating mount, a soundbar above it, and a markerboard alongside, with exposed cabling and a wall-mounted contr



**EX-08 · AV · 409:** Mobile computer cart labeled for Math & Computer Science tethered by a single USB cable to a wall box, beside a wall-mounted flat panel with loose cabling. Shows reliance on a





**EX-09 · AV · Room 407 (Math & Soldering lab):** Room 407 overview: rows of blue pegboard workbenches and stools beneath a recessed ceiling cassette AC unit and a ceiling-mounted projector, with two wall-mounted glass whiteb



**EX-10 · window · Classroom / exterior window:** Window head/jamb corner showing chalky deposit and streaking at the glazing perimeter where the seal meets the frame. Documents the window-reseal need and moisture/efflorescen

## 7 · Photo Exhibit Addendum — Part 2 of 2



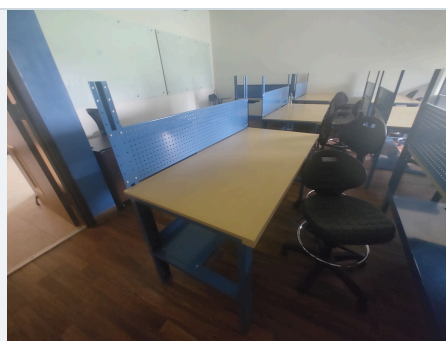
**EX-11 · window · Unspecified classroom/lab (dep:** Window head detail showing failed glazing: pronounced interpane fogging across roughly half the pane and a marginal top seal against bare cinderblock, documenting the window r



**EX-12 · window · 409:** Louvered wall grille over a window opening with bowed, misaligned blades and daylight leaking through gaps. Documents a damaged/ill-fitting louver assembly and a fenestration



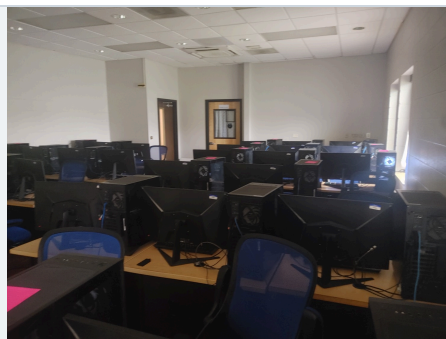
**EX-13 · bench · 407:** Room 407 lab casework: continuous black-topped bench with stainless sink, faucet, and wood cabinetry. Only a single surface-mounted countertop duplex receptacle serves this en



**EX-14 · bench · Room 407 (Math & Soldering lab:** Detail of Room 407 workbench assets: heavy blue steel benches with pegboard backboards, lower storage shelves, and matching wheeled stools. Clean documentation of reusable ben



**EX-15 · printer · Rooms 305/310 (Engineering):** Crisp close-up of the engineering 3D printers, clearly identifying the MakerBot Replicator and a second enclosed unit. Best documentation of the existing 3D-printer assets, wi



**EX-16 · printer · Engineering computer lab (Room:** Wide view of a fully equipped computer classroom: rows of tower desktops with dual flat-panel monitors on shared desks, blue task chairs throughout. Strong asset exhibit docum



**EX-17 · printer · Engineering computer lab (Room:** Companion angle of the computer classroom showing the full row layout against bare white cinderblock and wood-plank floor. Clearly documents the quantity of existing desktop w



**EX-18 · room-overview · Room 407 (Math & Soldering lab:** Corner of a bare-cinderblock room showing existing assets: a tall gray two-door metal storage cabinet (empty cabinetry), stacked chairs and folding-table legs, a rolling AV ca



**EX-19 · room-overview · Classroom (AI Studio candidate:** Wide overview of the candidate classroom: orange tablet-arm desks (one blue chair), a window with exterior greenery at left, a wall clock, and a dark exit door at right. A wal



**EX-20 · room-overview · Room 407 (Math & Soldering lab:** Room 407 full overview showing bench density, glass whiteboard, doorway to hallway, and a wall-mounted HVAC unit. Strong context shot of the empty candidate lab's overall cond

Remaining 51 exhibits (full set of 71) with room/condition explanations are on the online [resource & references page](#).

8 · Quote 1 — AI Studio + Engineering Lab (Title III grant-eligible)

|                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AI Studio · Rm 409</b><br>10–13 high-memory seats so students run local models: <b>6× Framework Desktop 128GB</b> (~\$2,851) + <b>4× Mac Studio M4 Max 64GB</b> (~\$2,600) + <b>1× DGX Spark</b> (\$4,699) + an <b>RTX 5090 workstation</b> + <b>Jetson AGX Thor</b> + Orin Nano kits + Synology NAS + rack UPS + an <b>85" demo wall</b> . Keep the dual smartboards/2 projectors. | <b>Math &amp; Soldering · Rm 407</b><br><b>5–10× Hakko FX-888DX</b> soldering stations + recirculating <b>BOFA 3D PrintPRO</b> -class fume/VOC extraction in the cabinetry/L-bend; bench instruments; an enclosed <b>Carbide Nomad 3</b> CNC. Reuses the existing blue pegboard benches + cabinetry. | <b>Engineering · Rm 305 + 310</b><br><b>≥4 Bambu Lab printers</b> : H2D large-format + <b>X1E</b> enterprise + <b>3× P1S + AMS</b> , enclosures, filament. <b>Repurpose the existing NVIDIA-GPU tower desktops</b> in 310 for AI hosting (10GbE NICs, RAM, IP-KVM) — engineering compute for near-zero new spend. |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Division                                                      | Representative real SKUs (June 2026)                                                                                        | Planning range           |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------|
| AI Studio compute (lean)                                      | Framework Desktop 128GB · Mac Studio M4 Max 64GB · DGX Spark · RTX 5090 WS · Jetson Thor/Orin · Synology DS1825+ · rack UPS | \$30,000–\$33,000        |
| Engineering Bambu fleet + GPU repurposing                     | Bambu H2D AMS Combo ~\$2,199 · X1E ~\$2,649 · 3× P1S+AMS ~\$799 · BOFA PrintPRO 3 · Intel X550-T2 NICs · PiKVM V4           | \$16,900–\$18,500        |
| Soldering bench + CNC                                         | 5–10× Hakko FX-888DX (~\$135) · bench fume extraction · Carbide Nomad 3 ~\$2,800                                            | \$8,000–\$9,000          |
| AV: 3 hallway display boards + 85" studio demo wall           | 3× ~50" commercial displays + UniFi Connect signage/media players + mounts · 85" demo wall ~\$4,600                         | \$8,250–\$9,120          |
| Furniture (movable) · Consumables/safety · Software/warranty  | Movable benches/chairs · filament/resin/PPE/flammables cabinet/ABC+Class-D extinguishers · licenses + extended warranty     | \$9,800–\$11,900         |
| <b>Quote 1 grant-eligible subtotal (equipment + supplies)</b> |                                                                                                                             | <b>\$88,242–\$98,820</b> |

**Display / project boards (your spec):** a board outside *each* of the three labs (409 / 407 / 305-310), auto-fed from in-room activity via UniFi Connect — "you see AI happening before you enter the room." The **85" demo wall** + per-semester access make the studio look intentional and impressive. **Grant tag:** items ≥\$2,000/unit are EQUIPMENT; the many <\$2,000 items are SUPPLIES — confirm with the grants officer whether rack components aggregate to a capitalized system.

## 9 · Quote 2 — Power / Electrical (Facility · NOT grant · subbed + GC-managed)

| Scope (lab-focused)                                                                                                                                          | Planning range    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Replace the FPE panel · 100A lab subpanel · 14 dedicated 20A circuits · 2× 30A L5-30 UPS circuits · conduit/raceway on cinderblock · lab touch-ups · permits | \$34,000–\$37,000 |

**Load calc (electrician verifies):** ~7–9kW continuous / ~12kW peak (~60–75A @120V) → a dedicated 100A subpanel is the floor (NEC 125% continuous derating; resolve 120/240V vs 208V; size the CNC circuit for inrush). **The Federal Pacific panel (EX-01-03) should be replaced regardless of the AI build** — a standalone life-safety item. **Named Huntsville subs to bid:** Madison Electric, Pro Electric, Lee Company, Blake Brothers, Urban Electric.

## 10 · Quote 3 — Network + Wi-Fi + Cameras + Access (Mixed: electronics grant / cabling facility)

| Division                                     | Real SKUs (June 2026)                                                                                                                      | Range             | Lane          |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------|
| UniFi core + pfSense                         | UniFi EnterpriseXG-24 (\$1,299) · 2× USW-Pro-Max-24-PoE (\$799) · UDM-Pro-Max (\$599) · <b>Netgate 6100 MAX pfSense</b> (\$999) · 42U rack | \$3,837–\$4,500   | Grant         |
| UniFi Protect cameras                        | UNVR-Pro (\$499) + 8TB HDDs · 10–12× G5/G6 Turret/Bullet (\$129–\$199) · AI Pro (\$499) for the print room                                 | \$3,055–\$3,800   | Grant         |
| UniFi Access — 4 doors, rotating credentials | 4–5× UA-Hub-Door (\$199) · G3 Reader Pro (\$379) + G3 Readers (\$159) · fail-safe HES strikes/maglock + REX + DPS + install                | \$8,400–\$9,000   | Equip+install |
| Structured cabling — lab-focused (SUBBED)    | Cat6A drops + patch panel + MDF rack rebuild (cabling labor = facility)                                                                    | \$12,000–\$14,000 | Facility      |

**Per-semester rotating credentials** are native to UniFi Access (PIN expiry, NFC deactivation, mobile/Wallet revoke). **Named Huntsville subs:** low-voltage — Travel Tech (256.970.4558), Interweave (256-837-2300), Heart Technologies (256-859-4000), State Systems; access — AAC Systems, Clear Link, LOCKTEC (256-881-9300), Tri-State Electrical. **ADA/egress:** all locks fail-safe; maglocks tie to the fire panel with REX + push-to-exit; Huntsville AHJ sign-off; low-voltage permit carried in Permits.



# 11 · Execute-Ready Invoice — Recommended Execution

Invoice basis: IDC (Jonathan Isaac) as General Contractor. Grant-eligible lines may be procured under Title III; facility lines are subbed and carry GC coordination. Re-quote each line to a sealed bid before purchase.

| #                                                             | Line item                                                                  | Amount (range)      |
|---------------------------------------------------------------|----------------------------------------------------------------------------|---------------------|
| I · Grant-Eligible Equipment & Supplies (Title III)           |                                                                            |                     |
| 1                                                             | AI Studio compute — studio/lean tier                                       | \$30,000–\$33,000   |
| 2                                                             | Engineering Bambu fleet + GPU-PC repurposing                               | \$16,900–\$18,500   |
| 3                                                             | Soldering bench + CNC                                                      | \$8,000–\$9,000     |
| 4                                                             | Consumables / safety / storage                                             | \$3,000–\$3,800     |
| 5                                                             | AV — 3 hallway display boards + 85" studio demo wall                       | \$8,250–\$9,120     |
| 6                                                             | UniFi network core + pfSense edge                                          | \$3,837–\$4,500     |
| 7                                                             | UniFi Protect cameras — priority labs + key hallway                        | \$3,055–\$3,800     |
| 8                                                             | UniFi Access — 4 doors, per-semester rotating credentials                  | \$8,400–\$9,000     |
| 9                                                             | Furniture (movable lab/studio)                                             | \$5,000–\$5,800     |
| 10                                                            | Software / licensing / extended warranty                                   | \$1,800–\$2,300     |
| Grant-eligible subtotal                                       |                                                                            | \$88,242–\$98,820   |
| II · Facility — Capital / Donor (NOT grant-eligible · SUBBED) |                                                                            |                     |
| 11                                                            | Power / electrical — FPE replacement, subpanel, circuits, raceway, permits | \$34,000–\$37,000   |
| 12                                                            | Structured cabling — lab-focused (Cat6A drops, patch, MDF rebuild)         | \$12,000–\$14,000   |
| 13                                                            | Raised access floor — AI Studio only (~300–400 sqft) + light room mod      | \$8,000–\$10,000    |
| 14                                                            | Permits / insurance / freight                                              | \$2,400–\$2,800     |
| Facility subtotal                                             |                                                                            | \$56,400–\$63,800   |
| Direct cost (I + II)                                          |                                                                            | \$144,642–\$162,620 |
| III · GC Coordination, Contingency & IDC Fee                  |                                                                            |                     |
| 15                                                            | GC coordination markup — 3–7% on the subbed facility base only             | \$1,692–\$4,466     |
| 16                                                            | Contingency — 6–8% on (direct + markup); 2026 memory-price volatility      | \$8,780–\$13,367    |
| 17                                                            | IDC Orchestration Fee — 10% of pre-fee subtotal                            | \$15,511–\$18,045   |
| ALL-IN TOTAL · Recommended commitment \$172,000               |                                                                            | \$170,625–\$198,498 |

### Payment schedule (proposed)

- **\$68,000 mobilization** — on signature; initiates procurement & sub scheduling.
- **~50% progress** — at equipment delivery + electrical/cabling rough-in.
- **Balance at completion** — commissioning, demo day, & closeout docs.

Pay deposit: [IDC Stripe payment link](#) (\$68,000)

### Acceptance & signatures

Signature-ready; DocuSign / SignWell / Adobe / print-scan accepted.

#### Provider

Jonathan Isaac  
Island Development Crew LLC  
[jon-isaac@islanddevcrew.com](mailto:jon-isaac@islanddevcrew.com)

Signature / Date

#### Client

Dr. Shushannah Smith  
Oakwood University · Math & CS  
[ssmith@oakwood.edu](mailto:ssmith@oakwood.edu)

Signature / Date

12 · Curriculum Spine — 2084 / John Lennox (responsible-AI ethics for an Adventist HBCU)

Textbook: "2084 and the AI Revolution" (Updated & Expanded Ed., John C. Lennox, Zondervan Reflective 2024) — chosen over the 2020 original because it engages the generative-AI/LLM era. Companion: the 2084 13-session Video Study for a flipped classroom. Two interleaved courses share one ethics spine; every 1–2 weeks ships one of 8 portfolio projects ("students don't study AI — they ship it").

| Wk    | Ethics seminar (Lennox)          | AI Engineering (Rm 409/310)                             | Project shipped              |
|-------|----------------------------------|---------------------------------------------------------|------------------------------|
| 1     | Ch.1 Mapping the territory       | Onboarding; Python/Git; local vs cloud LLM              | — setup                      |
| 2     | Ch.2–3 Origins / surveillance    | RAG chatbot on a local model (Rm 310 GPUs)              | P1 Local RAG assistant       |
| 3–4   | Ch.4–5 Narrow AI: promise & bias | CV classifier + bias/fairness audit; model cards        | P2 CV model + bias audit     |
| 5     | Ch.6 Upgrading humans            | Voice agent; assistive-tech use case + fabricate device | P3 Assistive-tech AI device  |
| 6–7   | Ch.7–8 AGI risk / personhood     | Agentic workflow + guardrails; multimodal app deploy    | P4 Agentic app w/ guardrails |
| 8–9   | Ch.9 Origin of moral sense       | Eval harness; red-team; responsible-AI write-up         | P5–P6 Eval + governance memo |
| 10–12 | Ch.10–13 Homo Deus / Revelation  | Capstone: ship a campus tool (à la Campus Companion)    | P7–P8 Capstone + demo day    |

Pricing model

- **All-in student price — \$300–\$350 / student:** a single combined price covering the **digital pass** (lab access, local-AI toolchain, project templates, eval harness, demo-board publishing, portfolio site) **and all course materials, including the 2084 textbook** — no separate book charge.
- Two new courses: **AI Engineering + hands-on Machining/CNC** — reduces required external co-op hours.

Why it lands at Oakwood

As the only SDA HBCU, Oakwood can teach AI with a **worldview-grounded ethics voice** secular schools struggle to articulate — data dignity, human-in-the-loop, Sabbath as a model for digital rest. Graduates leave with a portfolio *and* a point of view.

### 13 · Fundraising, Match Funds & the Naming Opportunity

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Grant &amp; corporate funding</b> <ul style="list-style-type: none"><li>• <b>Federal HBCU + AI:</b> NSF ExpandAI, NSF HBCU-EiR, NSF HBCU-UP, Dept of Ed Title III Part B (Oakwood is an active recipient).</li><li>• <b>Compute/hardware programs:</b> NVIDIA Academic Grant, AMD University Program, Google.org / UNCF.</li><li>• <b>Mission-fit philanthropy:</b> John Templeton Foundation, Lilly Endowment (faith &amp; AI), Mellon ("Unruly Intelligences").</li><li>• <b>Corporate matching-gift</b> programs.</li></ul> | <b>Notoriety &amp; the naming track</b> <ul style="list-style-type: none"><li>• <b>Voices to engage:</b> Dr. James Tour, John Lennox, Wesley Huff — credible on AI + faith + science.</li><li>• <b>Channels:</b> Modern Wisdom, Valuetainment, Diary of a CEO — to spotlight an HBCU building responsible AI.</li><li>• <b>Naming opportunity:</b> "The John Lennox AI Innovation Lab" (pending Oakwood approval) — pairs the 2084 curriculum with a named-gift envelope that funds the grant-ineligible facility scope.</li></ul> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 14 · Department-Wide Vision (donor / naming ask · \$245K–\$330K)

| Division (full scope)                                           | Range         | Division (full scope)                          | Range               |
|-----------------------------------------------------------------|---------------|------------------------------------------------|---------------------|
| AI Studio compute (pooled · RTX PRO 6000)                       | \$42K–\$50K   | Power / electrical (full dept)                 | \$34K–\$48K         |
| Engineering Bambu fleet (full)                                  | \$20K–\$23.9K | Structured cabling (80 drops + fiber)          | \$24K–\$36K         |
| Soldering + CNC (full)                                          | \$11K–\$14K   | Raised floor + Room-405 bathroom + room mod    | \$45.6K–\$62K       |
| Cameras (dept-wide) + Access (5 doors)                          | \$15K–\$19K   | Network core, AV, furniture, software, permits | \$24K–\$30K         |
| Department-Wide Vision — all-in (incl. GC markup + 10% IDC fee) |               |                                                | \$245,000–\$330,000 |

## 15 · Primary Source Links & References

Every quoted price traces to a named vendor or published benchmark (current June 2026). Re-quote each line to a sealed bid before purchase; 2026 memory-market pricing is time-sensitive.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AI / Compute</b> <ul style="list-style-type: none"><li>• <a href="#">Oakwood AI Lab — final web reference</a></li><li>• <a href="#">NVIDIA DGX Spark — marketplace</a></li><li>• <a href="#">NVIDIA DGX Spark — product</a></li><li>• <a href="#">Framework Desktop</a></li><li>• <a href="#">Apple Education Institution Price List</a></li><li>• <a href="#">NVIDIA Jetson Orin Nano Super</a></li><li>• <a href="#">NVIDIA Jetson AGX Thor</a></li><li>• <a href="#">Synology DiskStation</a></li></ul> | <b>Engineering / Fabrication</b> <ul style="list-style-type: none"><li>• <a href="#">Bambu Lab P1S</a></li><li>• <a href="#">Bambu Lab H2D (large format)</a></li><li>• <a href="#">Bambu Lab X1E (enterprise)</a></li><li>• <a href="#">Original Prusa XL</a></li><li>• <a href="#">Hakko FX-888DX soldering station</a></li><li>• <a href="#">Hakko fume extraction</a></li><li>• <a href="#">Donaldson BOFA 3D PrintPRO</a></li><li>• <a href="#">Carbide 3D Nomad 3 CNC</a></li></ul> | <b>Network / Security</b> <ul style="list-style-type: none"><li>• <a href="#">Ubiquiti UniFi store</a></li><li>• <a href="#">UniFi U7 Pro Max AP</a></li><li>• <a href="#">UniFi EnterpriseXG-24 (10GbE core)</a></li><li>• <a href="#">UniFi Dream Machine Pro Max</a></li><li>• <a href="#">Netgate 6100 MAX (pfSense+)</a></li><li>• <a href="#">UniFi Protect cameras</a></li><li>• <a href="#">UniFi Access (door control)</a></li></ul> |
| <b>Facility benchmarks</b> <ul style="list-style-type: none"><li>• <a href="#">Ethernet / structured cabling install cost</a></li><li>• <a href="#">Dedicated circuit install cost</a></li><li>• <a href="#">Raised access floor cost guide</a></li></ul>                                                                                                                                                                                                                                                     | <b>Curriculum</b> <ul style="list-style-type: none"><li>• <a href="#">2084 and the AI Revolution (Lennox, 2024)</a></li></ul>                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### Named Huntsville subcontractors (for the facility scope)

| Trade                              | Named Huntsville-area subcontractors (solicit 3 sealed bids each)                                                                                  |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Electrical                         | Madison Electric Inc · Pro Electric Inc · Lee Company · Blake Brothers · Urban Electric LLC                                                        |
| Low-voltage / cabling              | Travel Tech (256.970.4558) · Interweave Technologies (256-837-2300) · Heart Technologies (256-859-4000) · State Systems Inc.                       |
| Access control / locks             | AAC Systems LLC · Clear Link Systems · LOCKTEC Inc. (256-881-9300) · Tri-State Electrical Contractors                                              |
| General contracting / raised floor | Regional Builders (256-684-8833) · Wilcock Builders · Lee Builders · Pro Access Floors · Southwest Solutions Group · Intertech Commercial Flooring |

Full audited bill of materials with per-line SKUs and source URLs is in the package price book; all 71 walkthrough photographs at full resolution with explanations are on the online [resource page](#).

## 16 · Orchestration, Reporting & Next Steps

### Single point of accountability

IDC engages and schedules every vendor & Huntsville sub, manages the grant vs. facility funding lanes, and owns the timeline. One markup, one fee, one invoice.

### Live progress & reporting

A live HTML **progress dashboard** link is attached to the project email; **weekly & monthly** progress reports keep faculty & administration abreast through commissioning.

### Protected investment

Department cameras + per-semester rotating access control + cabling/power done to code (incl. FPE panel replacement) mean the equipment is controlled, managed, and protected from day one.

## Open items to confirm before sealed bids

- Final room dimensions & seat count (Rm 409 measured pass) — required for the raised-floor quote.
- Campus-IT vendor ruling: UniFi/pfSense allowed, or map to Aruba/Cisco/Extreme? Does access tie to the campus card?
- Oakwood tax-exempt status + grant eligibility ruling (setup labor, software/SaaS, networking hardware, cabling labor, access hardware).
- Tier to commit: Recommended ~\$172K vs. a sub-\$170K trim (single-lab cameras / 2-door access).

**Recommended next step.** Approve the Recommended Execution tier (~\$172,000 all-in), release the \$68,000 mobilization deposit (Stripe link, p.1 / p.11), and authorize IDC to (1) commission the licensed-electrician load study and (2) issue three sealed sub bids per trade. IDC returns firm, signed numbers within ~10 business days and begins the 90-day build.

Online: [Package hub](#) · [All 71 photos + references](#) · [Live dashboard](#) · [Pay \\$68K deposit](#)

### Authorized to proceed — Provider

Jonathan Isaac · Island Development Crew LLC

\_\_\_\_\_  
Signature / Date

### Authorized to proceed — Client

Dr. Shushannah Smith · Oakwood University

\_\_\_\_\_  
Signature / Date